Lab Exercise 12 - Start and Access Kubernetes Dashboard

Objective

To enable Kubernetes in Docker Desktop, deploy the Kubernetes Dashboard, and access it securely using a web browser on Windows.

Prerequisites

- Windows 10 / 11
- Docker Desktop installed
- Docker Desktop Kubernetes enabled
- Internet connection
- kubectl (comes bundled with Docker Desktop)

Step 1: Enable Kubernetes in Docker Desktop

1. Open **Docker Desktop**
2. Go to **Settings**
3. Select **Kubernetes**
4. Check **Enable Kubernetes**
5. Click **Apply & Restart**

Wait until Kubernetes status shows **Running** (green).

Step 2: Verify Kubernetes Cluster

Open **PowerShell** or **Command Prompt** and run:

- kubectl version --client
- Check cluster status:
- kubectl cluster-info

Check nodes:

```
kubectl get nodes
```

Expected output:

Node status should be **Ready**

Step 3: Deploy Kubernetes Dashboard

Apply the official Kubernetes Dashboard manifest:

```
kubectl apply -f

https://raw.githubusercontent.com/kubernetes/dashboard/v2.7.0/aio/deploy/recommended.yaml
```

Verify namespace creation:

```
kubectl get ns
```

You should see:

```
kubernetes-dashboard
```

Step 4: Verify Dashboard Pods

Check dashboard pods:

```
kubectl get pods -n kubernetes-dashboard
```

Expected status:

Running

Step 5: Create Admin User for Dashboard Access

Create a service account:

```
kubectl create serviceaccount dashboard-admin  -n kubernetes-dashboard
```

Create cluster role binding:

```
kubectl create clusterrolebinding dashboard-admin-binding --clusterrole=cluster-admin --serviceaccount=kubernetes-dashboard:dashboard-admin
```

Step 6: Generate Dashboard Login Token

Run the following command to get the token:

```
kubectl -n kubernetes-dashboard create token dashboard-admin
```

Copy the generated token (you will paste it in the browser later).

Step 7: Start Kubernetes Dashboard

Run the proxy command:

```
kubectl proxy
```

Keep this terminal **running**.

Step 8: Access Kubernetes Dashboard in Browser

Open a web browser and paste the following URL:

```
http://localhost:8001/api/v1/namespaces/kubernetes-dashboard/services/https:kubernetes-dashboard:/proxy/
```

Step 9: Login to Dashboard

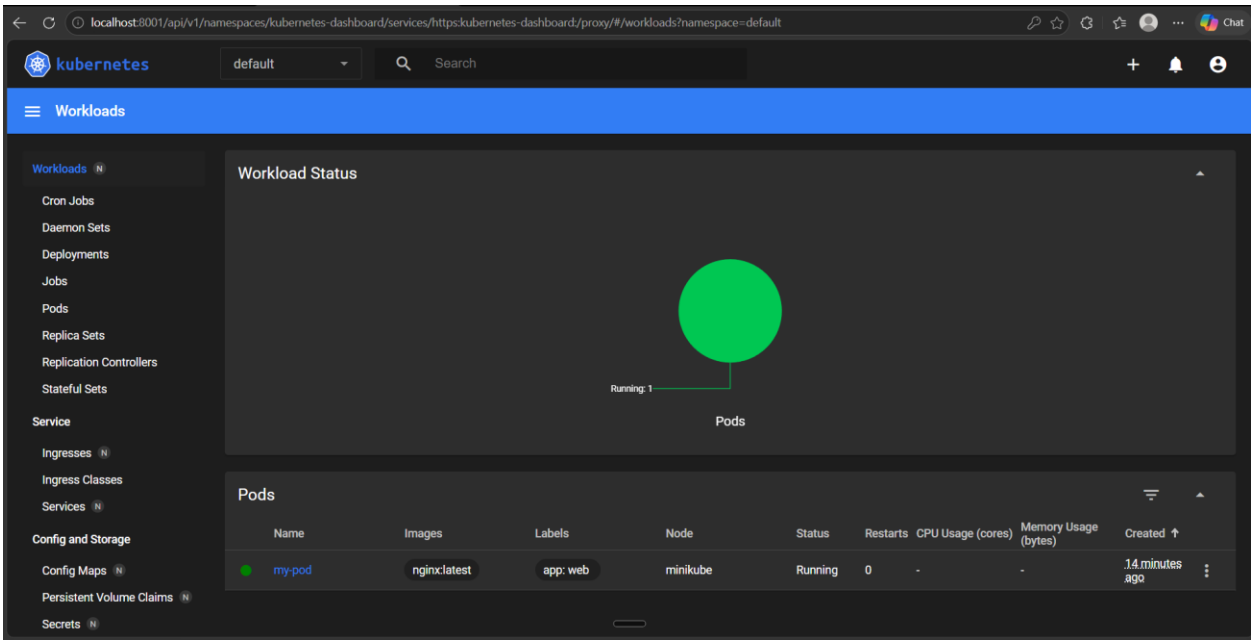
1. Select **Token** authentication
2. Paste the token generated earlier
3. Click **Sign In**

You should now see the **Kubernetes Dashboard UI**.

Step 10: Explore Dashboard

You can now view:

- Nodes
- Pods
- Deployments
- Services
- Namespaces
- ConfigMaps and Secrets




```
C:\Users\NK>kubectl cluster-info
Kubernetes control plane is running at https://127.0.0.1:61794
CoreDNS is running at https://127.0.0.1:61794/api/v1/namespaces/kube-system/services/kube-dns:dns/proxy

To further debug and diagnose cluster problems, use 'kubectl cluster-info dump'.

C:\Users\NK>]kubectl get nodes
']kubectl' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\NK>kubectl get nodes
NAME       STATUS   ROLES           AGE   VERSION
minikube   Ready    control-plane   13m   v1.35.1

C:\Users\NK>kubectl apply -f https://raw.githubusercontent.com/kubernetes/dashboard/v2.7.0/aio/deploy/recommended.yaml
namespace/kubernetes-dashboard created
serviceaccount/kubernetes-dashboard created
service/kubernetes-dashboard created
secret/kubernetes-dashboard-certs created
secret/kubernetes-dashboard-csrf created
secret/kubernetes-dashboard-key-holder created
configmap/kubernetes-dashboard-settings created
role.rbac.authorization.k8s.io/kubernetes-dashboard created
clusterrole.rbac.authorization.k8s.io/kubernetes-dashboard created
rolebinding.rbac.authorization.k8s.io/kubernetes-dashboard created
clusterrolebinding.rbac.authorization.k8s.io/kubernetes-dashboard created
deployment.apps/kubernetes-dashboard created
service/dashboard-metrics-scraper created
deployment.apps/dashboard-metrics-scraper created

C:\Users\NK>kubectl get ns
NAME                   STATUS   AGE
default                Active   13m
kube-node-lease        Active   13m
kube-public            Active   13m
kube-system            Active   13m
kubernetes-dashboard   Active   5s

C:\Users\NK>kubectl get pods -n kubernetes-dashboard
```

```
NAME                   STATUS   AGE
default                Active   13m
kube-node-lease        Active   13m
kube-public            Active   13m
kube-system            Active   13m
kubernetes-dashboard   Active   5s

C:\Users\NK>kubectl get pods -n kubernetes-dashboard
NAME                                         READY   STATUS              RESTARTS   AGE
dashboard-metrics-scraper-8d46b45f6-dnr8m   0/1     ContainerCreating   0          7s
kubernetes-dashboard-b44857bbb-x5w5w        0/1     ContainerCreating   0          7s

C:\Users\NK>kubectl create serviceaccount dashboard-admin -n kubernetes-dashboard
serviceaccount/dashboard-admin created

C:\Users\NK>kubectl create clusterrolebinding dashboard-admin-binding --clusterrole=cluster-admin --serviceaccount=kubernetes-dashboard:dashboard-admin
clusterrolebinding.rbac.authorization.k8s.io/dashboard-admin-binding created

C:\Users\NK>kubectl -n kubernetes-dashboard create token dashboard-admin
eyJhbGciOiJSUzI1NiIsImtpZCI6IlVSaDk5SmN5cWs4czdJbjJLYXNmcm9IQWRrVXNNZVdmX2tTMVNWWlhuVjgifQ.eyJhdWQiOlsiaHR0cHM6Ly9rdWJlcm5ldGVzLmRlZmF1bHQuc3ZjLmNsdXN0ZXIu
G9jYWwiXSwiZXhwIjoxNzc3MzUwNjIxLCJpYXQiOjE3NzczNDcwMjEsImlzcyI6Imh0dHBzOi8va3ViZXJuZXRlcy5kZWZhdWx0LnN2Yy5jbHVzdGVyLmxvY2FsIiwianRpIjoiZDM4ZDYyMjAtODQ1My00
DYyLTljNDUtOWU2M2ZjYTQwNmQ5Iiwia3ViZXJuZXRlcy5pbyI6eyJuYW1lc3BhY2UiOiJrdWJlcm5ldGVzLWRhc2hib2FyZCIsInNlcnZpY2VhY2NvdW50Ijp7Im5hbWUiOiJkYXNoYm9hcmQtYWRtaW4i
CJ1aWQiOiI2Nzg4MWU0MS00YTQ5LTRjZmYtODM0OC03YWZkZmIwNjNlOTcifX0sIm5iZiI6MTc3NzM0NzAyMSwic3ViIjoic3lzdGVtOnNlcnZpY2VhY2NvdW50Omt1YmVybmV0ZXMtZGFzaGJvYXJkOmRh
2hib2FyZC1hZG1pbiJ9.SRW9C61JWkSeyS84lYQE2yeUdrngxloHjNxnJzDp1h-NSKGFrmOkJXWMnFS1nqTlYf8WQ_rCKAr_oDxtibbYRf_DNACMtMmlali43fJYj1V5KGf3l08lg-6zhA82bOOW9cloX39
-53rLrefLIPAxrHhQ3dVNH_zNgBj3We5HEo5GV6VlK-THFbDju2NS39a7E-w8ODnYfNwnRupBcCYhVMsL8JI8V-ePXV_V6vgvqc0n4s9MUfXjso8g1kcZtbNF4J1vpGUaYUtvZFT9YmbzLjfohfDK70gnGX
VVK-qD4Kzz9TdxjgRsTTm4iUoajE7XEiRPZDEFBr13hUWWZhhg

C:\Users\NK>kubectl proxy
Starting to serve on 127.0.0.1:8001
^C
C:\Users\NK>kubectl get pods -n kubernetes-dashboard
NAME                                         READY   STATUS    RESTARTS   AGE
dashboard-metrics-scraper-8d46b45f6-dnr8m   1/1     Running   0          101s
kubernetes-dashboard-b44857bbb-x5w5w        1/1     Running   0          101s

C:\Users\NK>kubectl get services -n kubernetes-dashboard
NAME                        TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)    AGE
dashboard-metrics-scraper   ClusterIP   10.104.234.58    <none>        8000/TCP   104s
kubernetes-dashboard        ClusterIP   10.107.230.127   <none>        443/TCP    105s
```